

PCAP Network Analysis & Incident Report

Executive Summary

- **Target PCAP File:** evidence (1).pcapng
- **Total Capture Packets:** 11,089 Packets
- **Attacker IP Address:** 192.168.245.128
- **Victim Server IP Address:** 192.168.245.130

Task 1: Traffic Overview

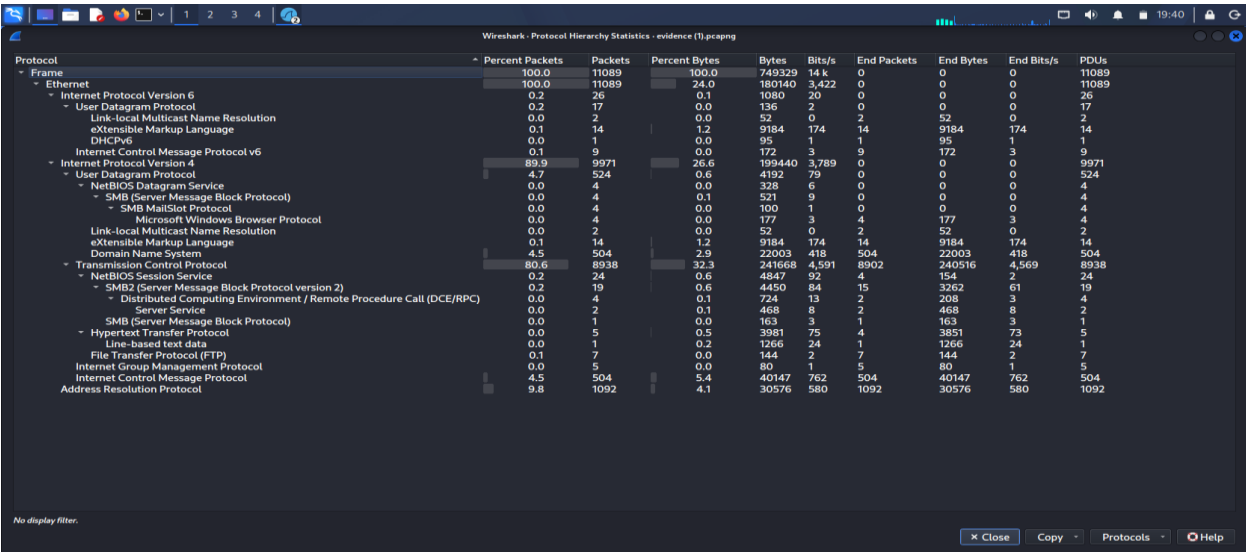
1. Protocol Distribution

The network capture contains a total volume of **11,089 packets**. Analysis of the structural layout reveals that the traffic profile is overwhelmingly dominated by **IPv4 User Datagram Protocol (UDP) Data**, accounting for **89.9%** of the total captured network footprint.

2. Service Profiles

Key application-layer protocols identified within the remaining packet architecture include:

- **File Transfer Protocol (FTP):** Consisting of 504 packets (4.5% of network volume).
- **Server Message Block (SMB/SMB2):** Used for network folder and file sharing interactions.
- **Hypertext Transfer Protocol (HTTP):** Facilitating unencrypted, clear-text web traffic.



Wireshark - Protocol Hierarchy Statistics - evidence (1).pcapng

Protocol	Percent Packets	Packets	Percent Bytes	Bytes	Bits/s	End Packets	End Bytes	End Bits/s	PDU/s
Frame	100.0	11089	100.0	749329	14 k	0	0	0	11089
- Ethernet	100.0	11089	24.0	180140	3,422	0	0	0	11089
- Internet Protocol Version 6	0.2	26	0.1	1080	20	0	0	0	26
- User Datagram Protocol	0.2	17	0.0	136	2	0	0	0	17
- Link-local Multicast Name Resolution	0.0	2	0.0	52	0	2	52	0	2
- eXtensible Markup Language	0.1	14	1.2	9184	174	14	9184	174	14
- DHCPv6	0.0	1	0.0	95	1	1	95	1	1
- Internet Control Message Protocol v6	0.1	9	0.0	172	3	9	172	3	9
- Internet Protocol Version 4	89.9	9971	26.6	199440	3,789	0	0	0	9971
- User Datagram Protocol	4.7	524	0.6	4192	79	0	0	0	524
- NetBIOS Datagram Service	0.0	4	0.0	328	6	0	0	0	4
- SMB (Server Message Block Protocol)	0.0	4	0.1	521	9	0	0	0	4
- SMB MailSlot Protocol	0.0	4	0.0	100	1	0	0	0	4
- Microsoft Windows Browser Protocol	0.0	4	0.0	177	3	4	177	3	4
- Link-local Multicast Name Resolution	0.0	2	0.0	52	0	2	52	0	2
- eXtensible Markup Language	0.1	14	1.2	9184	174	14	9184	174	14
- Domain Name System	4.5	504	2.9	22003	418	504	22003	418	504
- Transmission Control Protocol	80.6	8938	32.3	241668	4,591	8902	240516	4,569	8938
- NetBIOS Session Service	0.2	24	0.6	4847	92	4	154	2	24
- SMB2 (Server Message Block Protocol version 2)	0.2	19	0.6	4450	84	15	3262	61	19
- Distributed Computing Environment / Remote Procedure Call (DCE/RPC)	0.0	4	0.1	724	13	2	208	3	4
- Server Service	0.0	2	0.1	468	8	2	468	8	2
- SMB (Server Message Block Protocol)	0.0	1	0.0	163	3	1	163	3	1
- Hypertext Transfer Protocol	0.0	5	0.5	3981	75	4	3851	73	5
- Line-based text data	0.0	1	0.2	1266	24	1	1266	24	1
- File Transfer Protocol (FTP)	0.1	7	0.0	144	2	7	144	2	7
- Internet Group Management Protocol	0.0	5	0.0	80	1	5	80	1	5
- Internet Control Message Protocol	4.5	504	2.4	40147	762	504	40147	762	504
- Address Resolution Protocol	9.8	1092	4.1	30576	580	1092	30576	580	1092

Task 2: Identify Malicious Indicators

1. Host Discovery Probing

The initial segments of the packet capture demonstrate rapid, alternating ICMP Echo Requests (Pings) and DNS queries passing between the host machines. This baseline signature matches early-stage passive asset discovery and host scanning layout setups.

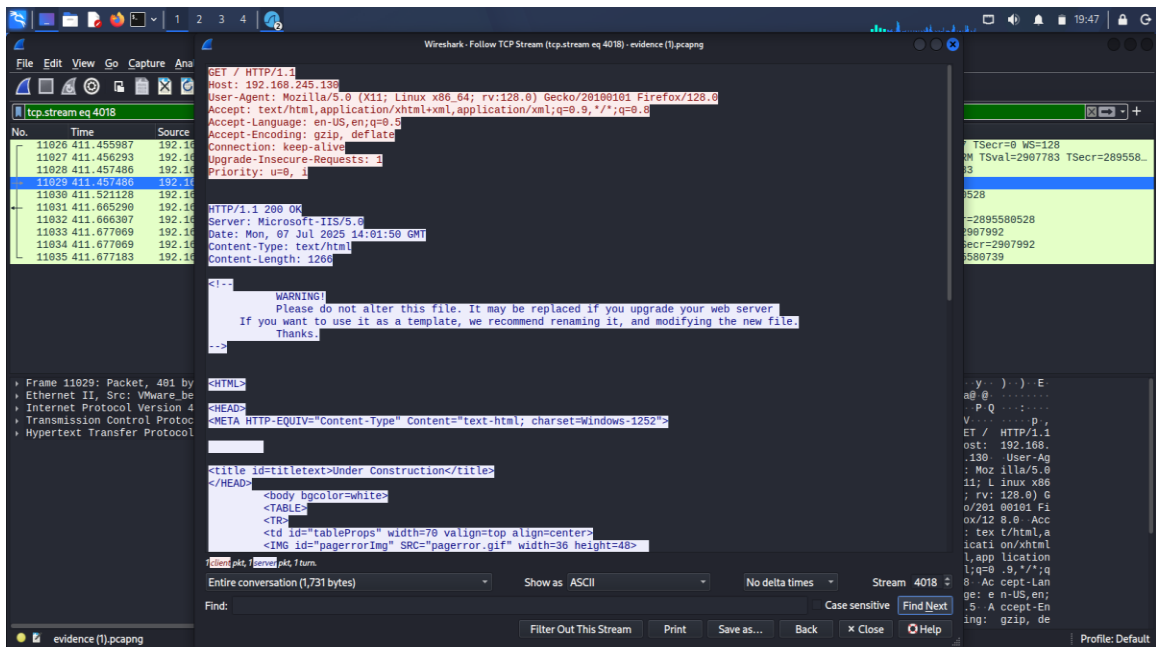
2. Authentication Spraying / Brute-Forcing

Applying targeted application filtering isolated a heavy concentration of automated connection requests targeting the legacy FTP service daemon. The rapid frequency of authentication attempts followed by explicit server rejection messages serves as a definitive indicator of an active brute-force or credential-guessing attack profile.

Task 3 & 4: Reconstruct Communication & Data Leakage

1. Plain-Text Web Exposure

Reconstruction of the unencrypted web traffic stream (TCP Stream 4018) indicates that the target server is hosting a web application running on an obsolete and highly vulnerable server software package: Microsoft-IIS/5.0.

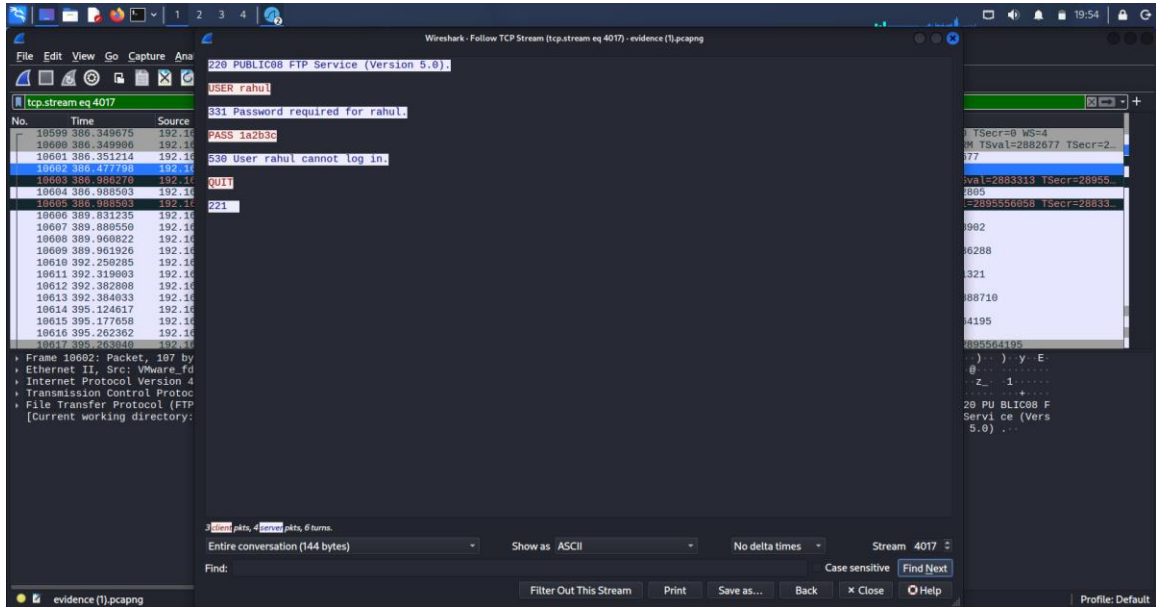


2. Credential Harvesting (FTP Analysis)

Because the services were deployed without cryptographic tunnel protection, the automated authentication sequence exposed sensitive security details in plain text over the wire. Under **TCP Stream 4017**, the following identity assets were cleanly harvested:

- **Exposed Target Username:** rahul
- **Exposed Target Password:** 1a2b3c

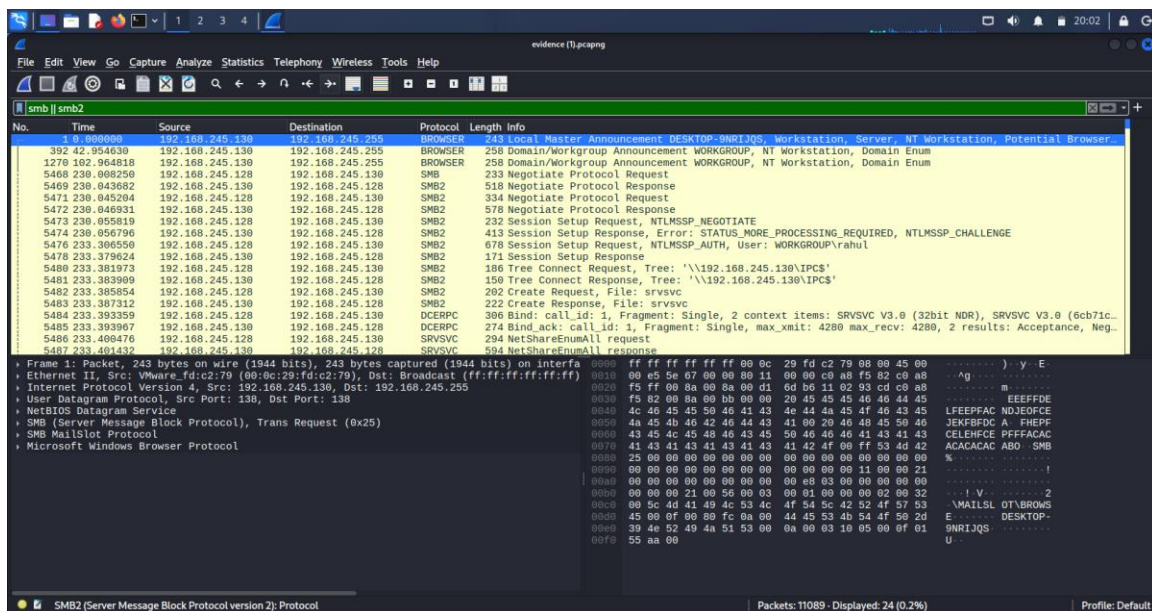
While the server dropped this specific session connection with an error flag (530 User rahul cannot log in), the attacker successfully captured a valid set of system credentials.



Task 5: Final Attack Scenario & Conclusion

1. Attack Timeline Sequence

1. **Reconnaissance:** The attacker (192.168.245.128) performed early network probing to find active hosts.
2. **Credential Sniffing/Guessing:** The attacker launched a brute-force sequence against the FTP service, harvesting the functional credentials for user account rahul.
3. **Lateral Movement / System Intrusion:** Armed with valid credentials, the attacker shifted away from the blocked FTP port and targeted the Windows file-sharing infrastructure (SMB/SMB2).
4. **Successful Compromise:** The attacker submitted an NTLM authentication request (NTLMSSP_AUTH) using the compromised account WORKGROUP\rahul. The server accepted the credentials, granting the attacker unauthorized remote terminal access to the administrative \\192.168.245.130\IPC\$ inter-process communication network share.



2. Mitigation Recommendations

- **Disable Deprecated Services:** Immediately decommission the legacy **Microsoft-IIS/5.0** and old FTP service versions.
- **Enforce Encryption:** Upgrade all plain-text web interfaces to **HTTPS (TLS)** and force file transfers over encrypted protocols like **SFTP**.
- **Account Remediation:** Force an immediate global password change for the user account **rahul** across the entire domain workspace environment.